

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the “B-” prefix indicate?
- (A) Between two tokens
 - (B) Beginning of an entity
 - (C) Inside an entity
 - (D) Outside any entity
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- (A) It uses only words (observations) and ignores labels
 - (B) It models transitions between labels but does not consider the words
 - (C) It models emissions $P(\text{word} \mid \text{label})$
 - (D) It is identical to an HMM
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- (A) $P(\text{label} \mid \text{previous label})$
 - (B) $P(\text{word} \mid \text{label})$
 - (C) $P(\text{label} \mid \text{word})$
 - (D) $P(\text{word} \mid \text{previous word})$
- Q4. In an n-gram language model, a bigram LM predicts the next word using:
- (A) No previous words
 - (B) The previous 1 word
 - (C) The previous 2 words
 - (D) The entire sentence history
- Q5. Which statement about perplexity is correct?
- (A) It's the raw probability of a sentence
 - (B) It's the inverse probability of a test text, normalized by length
 - (C) It's the fraction of correctly labeled tokens
 - (D) It's the number of unique words in the corpus

End of Paper